Conversion to Camphene to Camphor (Day 3)

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Introduction

In this three-day experiment the conversion of camphene to camphor occurred. In Day 3, the conversion of borneol to camphor is performed. For this part of the experiment the use of camphene, acetone, dichloromethane, acetic acid, aqueous solution of sodium bicarbonate, sodium hypochlorite, borneol, and camphor are being used to perform this experiment.

Materials and Methods

In this experiment it started by prepping with multiple steps. The first step was to put 0.3 g of the borneol that was made in the previous lab to a 10 mL round-bottom flask, then 0.3 mL of acetic acid was added to it. To dissolve the mixture 1 mL of acetone was added and then a. boiling chip. The next thing that was performed was the reflux with the borneol mixture for it to dissolve. After all of the borneol dissolved, 6 mL of aqueous sodium hypochloride solution was added dropwise for around 35 minutes. After each drop, the solution was swirled to make sure it was mixed thorough and then after the last drop the reaction was allowed to complete for 15 more minutes. After that was completed, the mixture was left to cool to room temperature and then the camphor was extracted with two portions (2 mL each) of methylene chloride. The combined methylene chloride was then neutralized with 2 mL of saturated sodium bicarbonate. Once the methylene chloride was neutralized, the organic solution was washed with 2 mL of 5% sodium bisulfite and 2 mL of water. The organic solution was then dried over anhydrous sodium sulfate and was put over a water bath to evaporate the DCM (dichloromethane).

Results

Borneol (g)	0.789
Watch Glass (g)	26.964

Camphor (g)	10.466
Percent Yield (%)	7.5%

Table. 1. Borneol Watch Glass, Camphor, Percent Yield

Calculations

$$\frac{0.789}{10.466} x100 \; = \; 7.5\% \; percent \; yield$$

Discussion

In this experiment, there were multiple sources of error. One main source was when the reflux was disassembled before it reached room temperature. This is important because if the solution is still hot during the washing it can mess up the solution. Another source of error is when the product had difficulty staying pure. There are a lot of extractions that occurred in the experiments which can lead to multiple different errors and can mess up the camphor in the end. The reaction that takes place is in water. One thing that was used that is very dangerous was DCM (dichloromethane). DCM is very dangerous to the environment and was handled with special care throughout the experiment.

References

2022 CHEM 2081 Organic Chemistry 2 Lab Manual. Auburn, (AL): Auburn University